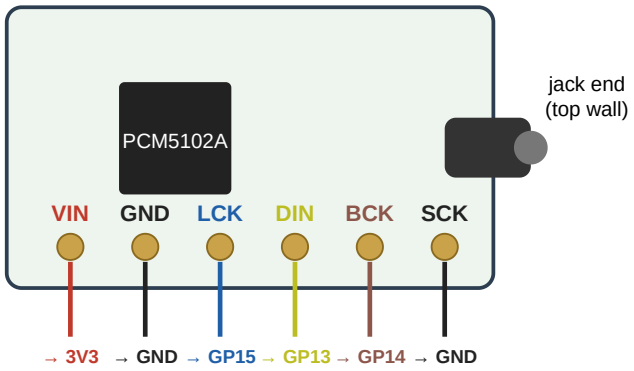
Pod — GY-PCM5102 DAC wiring

Replacement for the Adafruit board. Firmware pin map (sdk/common/pod_pins.h): **DIN = GP13 (pin 17), BCK = GP14 (pin 19), LCK/WSEL = GP15 (pin 20)**. Same three Pico pins as before, so the old wires 14–16 can be reused if they are still there; only the DAC end changes.

GY-PCM5102, seen from the component side

(go by the names printed on the board, not by position)



6-pin header: 5 wires to the Pico, SCK tied to GND
(the chip then makes its own master clock from BCK)

Pico end of each wire

GP0 row, last four pins at the antenna end
(perfboard row 8, columns Q R S T)

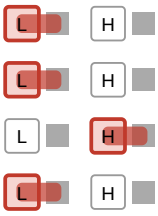




R10 = the screen socket's VCC hole (3V3 net);
Pico pin 36 (5th from the USB end) is the same net.
Silent on 3.3 V? Move VIN to VSYS (pin 39).

Back of the board: four solder bridges.

Without no. 3 the DAC is mute.



Each jumper: a centre pad with an H pad one side, an L pad the other, numbered 1–4.
A blob of solder across centre+L (centre+H for no. 3).
Boards often ship with 1, 2, 4 bridged and no. 3 open: check all four with a magnifier.

The six connections

VIN	→ 3V3 (Pico pin 36 / hole R10)
GND	→ GND (Pico pin 18 / hole R8)
LCK	→ GP15 (Pico pin 20 / hole T8)
DIN	→ GP13 (Pico pin 17 / hole Q8)
BCK	→ GP14 (Pico pin 19 / hole S8)
SCK	→ GND (short to the GND pin beside it)

Test with pod_dac_test.uf2 (beeps L / R / both), then pod.uf2.

Mounting for the Rev B case: top-left of the perfboard, board edge x 3.8–21.3 (the SD socket is to its right, the corner screw block to its left), jack end sticking **3.5 mm past the perfboard's top edge**, module about 4 mm above the perfboard so the jack centre is ~8 mm up. Easiest: solder a 6-pin male header into row 33 columns N–S (x 5–18), seat the DAC's header on it leaving the plastic spacer 1.5 mm proud, solder, trim. Tell me the measured perfboard-top-to-jack-centre height and I update *JACK_H* if it is not 8 ± 1.5.

Order of work:

1. Bridge the four pads on the back first (3 → H, the rest → L), with the board loose on the bench.
2. Short SCK to GND on the header (a solder bridge between the two adjacent pins, or one short wire).
3. Solder the five wires: VIN, GND, LCK, DIN, BCK, as above. Meter check: DAC GND beeps to Pico pin 18; DAC VIN reads 3.3 V with the Pico powered; no beep between VIN and GND.
4. Flash **pod_dac_test.uf2**, plug in headphones. Silence → recheck jumper 3 and that LCK/BCK/DIN are not swapped (the pin-finder firmware blinks GP13 at 1 Hz, GP14 fast, GP15 slow).